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Roll No

CE-6003-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Geotechnical Engineering - I

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume missing data suitably.

1. a) Define the terms: 4
i) void ratio
ii) specific gravity
b) Derive the relationship between void ratio (e), Degree of saturation (S_r), Water content (w) and specific gravity (G) of soil mass. 5
c) Describe the Indian standard soil classification system of soil. 5
2. a) Discuss three phase system of soil with proper sketch. 5
b) Discuss in brief about one dimensional theory of consolidation. 4
c) Discuss laboratory procedure as per IS code to find plastic limit of soil? 5
3. a) Derive the equation of stress distribution in soils according to Boussinesq. 7
b) Explain Mohr-coulomb's theory of shear failure of soil and derive equation of Mohr's stress circle. 7

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4. a) Explain procedure to find the shear strength of soil by triaxial compression test. 7
b) Explain about Newmark's influence chart with its sketch. 7
5. a) Explain about types of slope failures and stability analysis of earth dams. 7
b) Explain about Taylor's stability number and rotational slips. 7
6. a) A concentrated load of 22.5 kN acts on the surface of a homogeneous soil mass of large extent. Find the stress intensity at a depth of 15 meters and (i) directly under the load, and (ii) at a horizontal distance of 7.5 meters. Use Boussinesq's equations. 7
b) Explain about analytical and graphical methods of stability analysis of slopes. 7
7. a) An 6m high retaining wall with $\phi' = 36$ degrees, $c' = 0$, $\gamma = 15\text{kN/m}^3$. 7
i) The Rankine active force per unit length of the wall and the location of the resultant.
ii) The Rankine passive force per unit length of the wall and the location of the resultant.
b) Explain about Culmann's lateral earth pressure theories. 7
8. Discuss the following:
a) Reinforced earth retaining wall. 4
b) Active and passive earth pressure 3
c) Consistency limits 4
d) Pore water pressure 3

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